

The Monetary–Financial–Fiscal Model (MFFM)

A global projection and scenario tool —
one coherent “what-if” for 50 economies, and its policy implications.

Mátyás Farkas

International Monetary Fund

Senior Management briefing ■ *[date / venue]*

Views expressed are my own and do not necessarily represent the IMF.

Why now: the next generation of the Fund's global macro-financial model

Today: the GMF (Vitek)

- ▶ Served the Fund well as a global macro-financial workhorse.
- Not updated in estimation — *[MF/Jesper: specifics]*
- Missing FSAP / Article IV features — *[MF/Jesper: list the needs]*
- *[MF/Jesper: any further gaps]*

What surveillance now needs

- + Current estimates, refreshed each cycle.
- + Fast, transparent scenarios for FSAP & Article IV.
- + Genuine cross-border transmission, not country-by-country.
- + *[MF/Jesper: the specific FSAP/AIV features]*

The case:

[MF/Jesper: one-line motivation — e.g. “The MFFM supersedes the GMF: re-estimated, scenario-ready, and built for FSAP–Article IV work.”]

What the MFFM adds — five things it does that a static model cannot

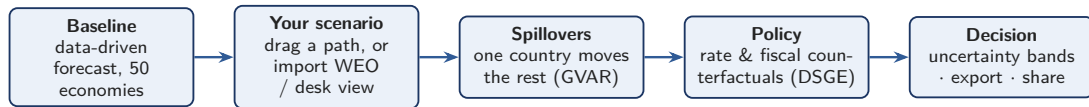
- + **Re-estimated and current.** Bayesian VAR baselines re-estimated on the latest data, for **50 economies** — one consistent methodology everywhere.
- + **Interactive scenarios.** Build a “what-if” by hand, or from an outside forecast, in minutes — the model keeps every variable mutually consistent.
- + **Real cross-border amplification.** A coupled *global* solve — a shock in one economy travels through trade and finance to the rest, not a sum of separate country runs.
- + **Uses what we already produce.** WEO, consensus and desk forecasts enter as *ingredients*, at a trust level you choose.
- + **Honest and reproducible.** Uncertainty bands throughout; a whole session — every country, every assumption — saved and shared as a single file.

In one line:

The same trusted models, made **current, interactive, global, and shareable.**

One tool: a coherent global scenario, its spillovers, and the policy trade-offs

A point-and-click platform for **50 economies** — the Fund's workhorse models (Bayesian VAR baselines, a cross-country GVAR, DSGE policy) behind a single interface. Today's tour, in five steps:



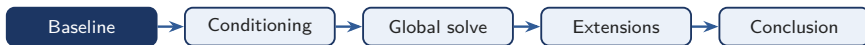
Why it matters

- + **Speed** — a coherent global “what-if” in an afternoon, not a forecasting round.
- + **Coherence** — every number is mutually consistent, at home *and* across borders.
- + **Reuse** — folds in the forecasts we already produce (WEO, SPF, desk).

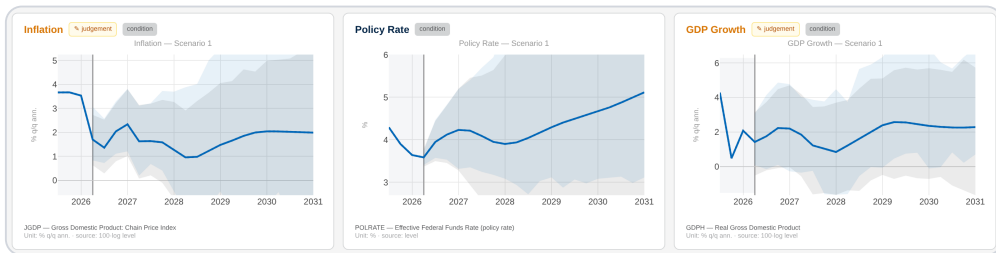
What you can ask it

- ▶ If US growth surprises — who else moves, and by how much?
- ▶ If the Fed holds rates higher for longer, what happens here?
- ▶ If we take the WEO as given, what does it imply for everything else?

① Start with the baseline — no inputs, the model's own read of every economy



No assumptions required. Each economy is a Bayesian VAR, linked into one global system.

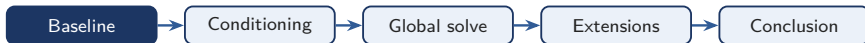


US inflation, policy rate and GDP — one consistent forecast, with uncertainty bands.

The baseline:

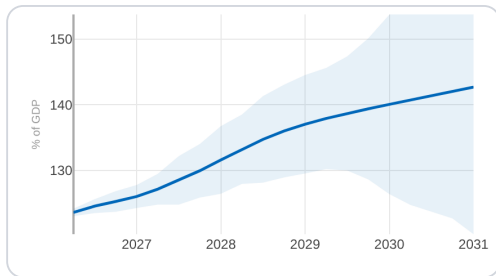
A ready-made, internally consistent forecast for all **50 economies** — the common starting point every scenario departs from.

① Fiscal accounting, built in — the debt path drops out of the same run



Revenue, spending, the interest bill and the deficit are projected **together** — and debt-to-GDP follows from the **budget identity**.

No separate spreadsheet: the fiscal story is **consistent with the macro forecast by construction**.

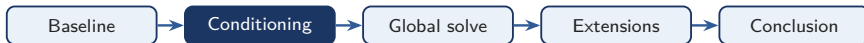


US debt-to-GDP rises to ~143% by 2031, with band.

Why it matters:

Debt sustainability drops out of the *same* run — the macro and fiscal stories can never contradict each other.

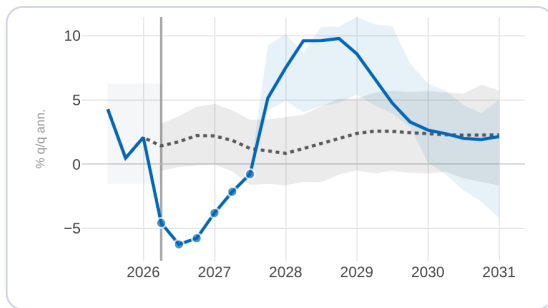
② Tell it two things you believe — it works out the rest, consistently



Recipe · Credit crunch

1. Pin **spreads jump**.
2. Pin **lending falls**.

Two pins in; the model works out a **full, coherent response** across every variable.

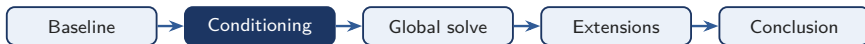


US GDP response to the two credit pins, with bands.

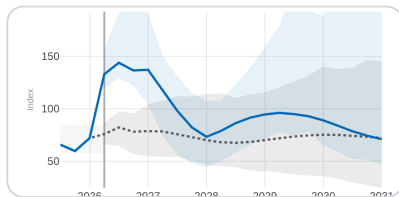
The power move:

A couple of believable pins → a full, consistent macro response — a structural shock, no econometrics.

② A richer story? Just add a few more pins — an energy shock

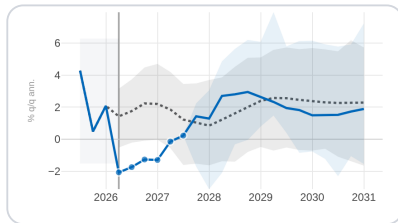


You pin



An oil-price path — the size and shape of the spike.

Out comes



US stagflation: **prices up, output down.**

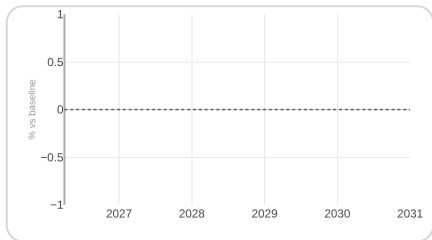
Fit the pins to the story:

The richer the story, the more you say up front — an oil path plus a few transmission pins, and the same move does the rest.

③ A shock abroad, a hit at home — the part a country-only read cannot see

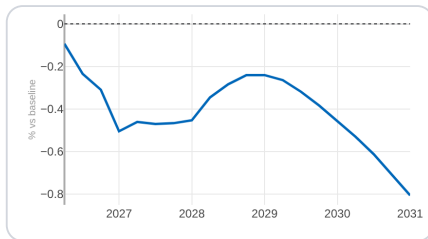


Country-only read



Euro-area GDP vs baseline: **flat**.

Global solve

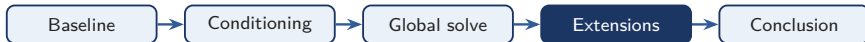


The same shock arrives: **-0.8pp** by 2031.

Why the tool earns its keep:

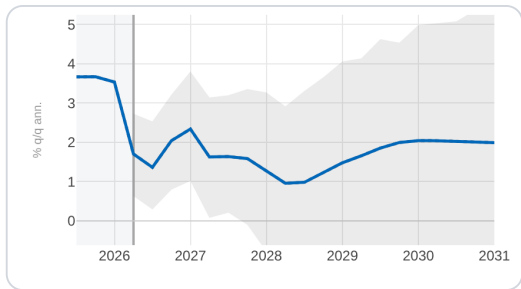
A country-only read — like the **old GMF** — shows **zero**; only the coupled solve reveals the hit.

④ Pin the destination — and the whole path tilts toward it



Know only **where things should end up** — the inflation target, potential growth, the neutral rate?

Set that **long-run anchor**; the model tilts the path toward it, gently, **without breaking the near term**.

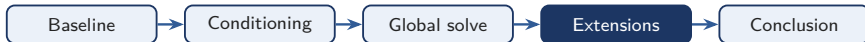


US inflation: the tail lands on the anchor (dashed); near term untouched.

Long-run anchors:

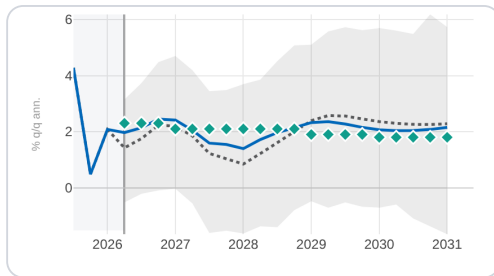
Set where a variable **settles**; the path re-shapes to reach it, near term intact.

④ Fold in an outside forecast as an ingredient — at a trust level you choose



A WEO number, a consensus figure, a desk view? **Import it directly** — as one more ingredient, not a hard override.

You set the **trust dial**: from “just a hint” to “pin it exactly”. Everything else stays **consistent** around it.

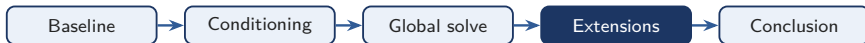


US GDP: teal diamonds are the imported WEO path.

External forecasts:

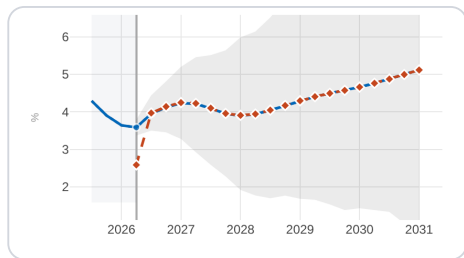
Reuse the WEO and desk forecasts we *already* produce — as inputs, at a **trust level you control**.

④ Ask “what if the central bank does X?” — a genuine policy counterfactual



This tells you **what a policy choice does** — a **structural** answer from a fully specified DSGE model.

Layer a **policy shock** on any scenario; the model traces the **counterfactual path**.



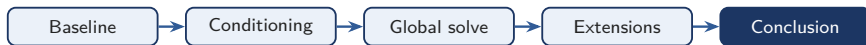
US policy rate: counterfactual vs. baseline scenario.

Policy counterfactuals:

From “what will happen” to “**what should we do**” — a structural (MMB/DSGE) policy experiment.

[on the roadmap: deeper MMB library, cross-country WEO propagation, external-balance/NFA satellite]

⑤ One tool now gives us a current, global, and shareable read on policy we ask you to adopt it



What we built, in three lines:

- + **Current & consistent** — every economy sits on the same up-to-date vintage, so the numbers add up across desks
- + **Genuinely global** — shocks travel across 50 economies through trade and finance, not one country at a time
- + **Our forecasts, built in** — WEO and staff views enter as dials, and any run is one-click **shareable and reproducible**

The arc: a live baseline → **condition** it on the story you care about → let it **propagate globally** → layer in **policy and external-balance** extensions.

The ask: *[MF/Jesper: endorse the tool, resource the last-mile hardening, and adopt it into the FSAP / Article IV workflow.]*

It is **live today**, covering **50 economies** ready to pilot on the next surveillance cycle.

The one line:

One live, shareable tool turns scattered forecasts into a single consistent, global answer to “what if?”.